

Plant biotechnology

BT421

Agrose gel electrophoresis

Agarose gel electrophoresis

Agarose gel electrophoresis is a method of gel electrophoresis used in biochemistry, molecular biology, genetics, and clinical chemistry to separate a mixed population of macromolecules such as DNA or proteins in a matrix of agarose, one of the two main components of agar.

Biomolecules are separated by applying an electric field to move the charged molecules through an agarose matrix, and the biomolecules are separated by size in the agarose gel matrix.

Agarose gel is easy to cast, has relatively fewer charged groups, and is particularly suitable for separating DNA of size range most often encountered in laboratories, which accounts for the popularity of its use

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. The separated DNA may be viewed with stain, most commonly under UV light, and the DNA fragments can be extracted from the gel with relative ease.

Most agarose gels used are between 0.7–2% dissolved in a suitable electrophoresis buffer.

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A number of factors can affect the migration of nucleic acids:

- 1- The dimension of the gel pores (gel concentration).
- 2- Size of DNA being electrophoresed.
- 3- The voltage used.
- 4- The ionic strength of the buffer.
- 5- The concentration of intercalating dye such as ethidium bromide if used during electrophoresis.

Smaller molecules travel faster than larger molecules in gel, and double-stranded DNA moves at a rate that is +inversely proportional to the logarithm of the number of base pairs.

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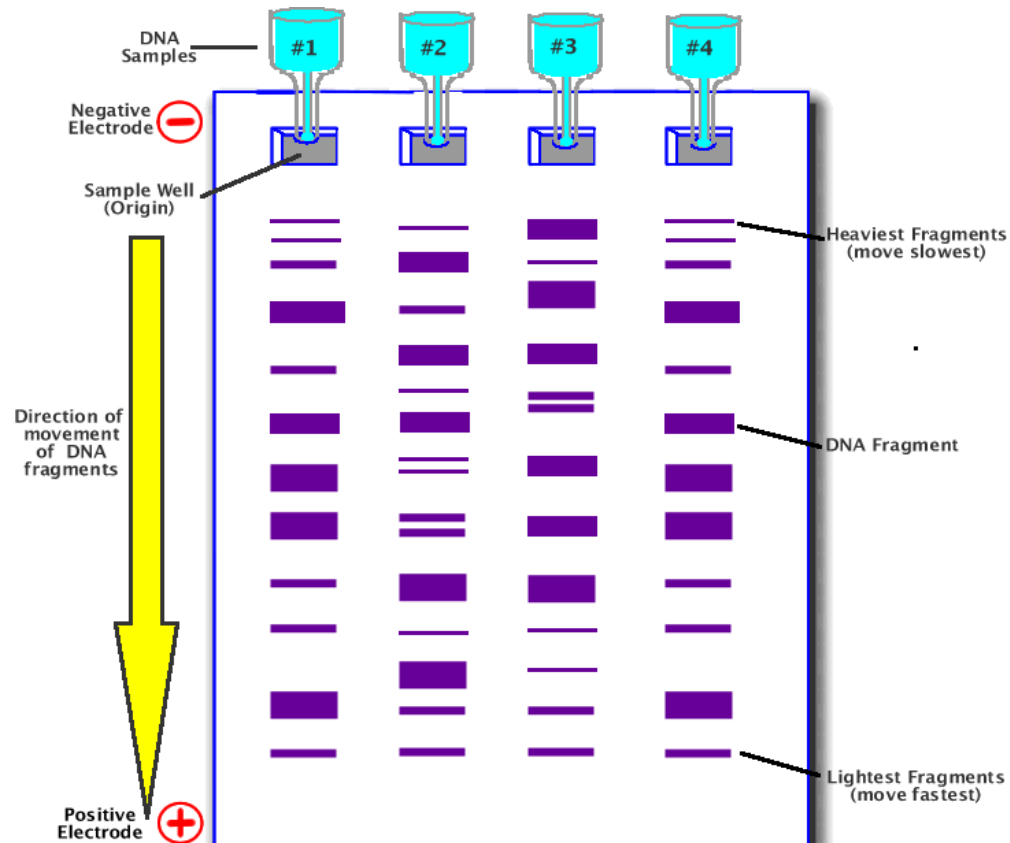
Mechanism of migration and separation

The negative charge of its phosphate backbone moves the DNA towards the positively charged anode during electrophoresis.

The migration of DNA molecules in solution, is independent of molecular weight during electrophoresis.

The gel matrix is therefore responsible for the separation of DNA by size during electrophoresis.

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Gel Electrophoresis
(Creating a DNA Profile)